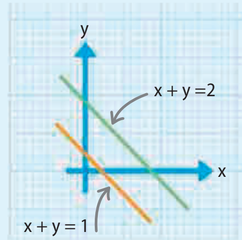
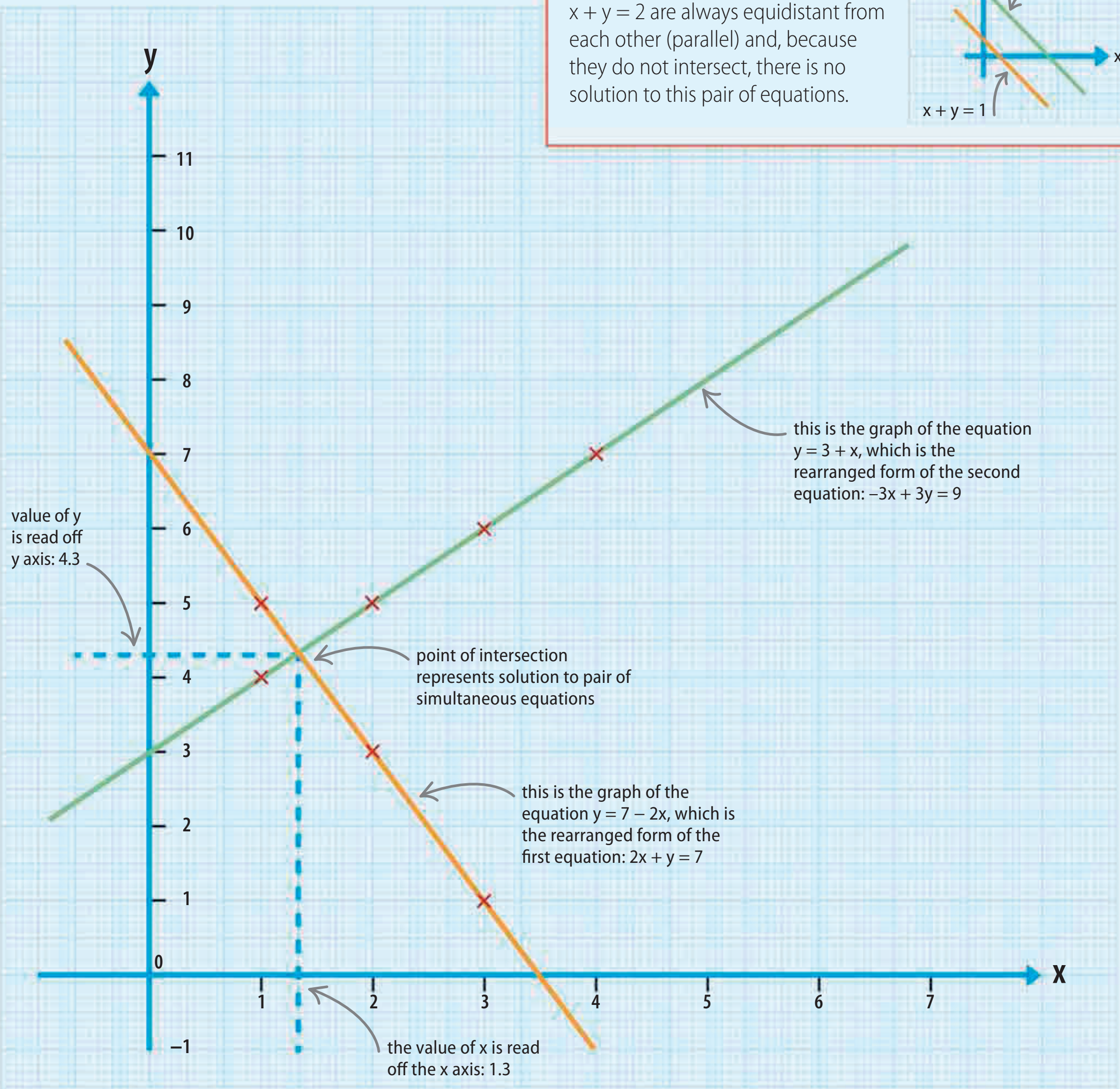
Draw a set of axes, then plot the two sets of x and y values. Join each set of points with a straight line, continuing the line past where the points lie. If the pair of simultaneous equations has a solution, then the two lines will cross.

LOOKING CLOSER

Unsolvable simultaneous equations

Sometimes a pair of simultaneous equations does not have a solution. For example, the graphs of the two equations $x + y = 1$ and $x + y = 2$ are always equidistant from each other (parallel) and, because they do not intersect, there is no solution to this pair of equations.





The solution to the pair of simultaneous equations is the coordinates of the point where the two lines cross. Read from this point down to the x axis and across to the y axis to find the values of the solution.

$$x = 1.3 \qquad y = 4.3$$